

Problem Statement

September 5, 2026

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Compute the derivative of the function $f(t) = t^2$ with respect to the variable t :

$$\frac{d}{dt} t^2.$$

Key Principles and Formulas

- **Definition of the derivative (limit form):** For a function $f(t)$,

$$\frac{df}{dt} = \lim_{h \rightarrow 0} \frac{f(t+h) - f(t)}{h}.$$

- **Algebraic expansion:** $(a+b)^2 = a^2 + 2ab + b^2$.
- **Power rule (derived from the limit definition):** $\frac{d}{dt} t^n = nt^{n-1}$ for any real n . In this problem we will derive the case $n = 2$ from first principles.

Detailed Step-by-Step Solution

$$\begin{aligned} \frac{d}{dt} t^2 &= \lim_{h \rightarrow 0} \frac{(t+h)^2 - t^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{t^2 + 2th + h^2 - t^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{2th + h^2}{h} \\ &= \lim_{h \rightarrow 0} (2t + h) \\ &= 2t. \end{aligned}$$

Final Answer

$$\boxed{2t}$$

Study Notes and Key Takeaways

- When applying the limit definition, always expand $(t+h)^2$ fully before canceling terms; missing the h^2 term leads to incorrect results.
- After canceling the common factor h , the remaining expression should be continuous at $h = 0$; this allows you to substitute $h = 0$ directly.
- The power rule $\frac{d}{dt}t^n = nt^{n-1}$ follows from this same process for any integer n ; mastering the $n = 2$ case builds intuition for the general rule.